Research gap and claim boundary

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The two communities of sec. 5 provide substantial pieces of the problem, but the cited threads do not answer the same question. The gap is not a claim that either field is incomplete; it is the untested intersection between active-inference belief sharing and robust generalized Bayes.

Research gap and claim... (part 2)

This section describes that intersection thread by thread, states the scoped bridge evaluated here, and records the evidence boundary that travels with it.

Five reviewed threads and their open intersection

Five threads in the reviewed literature run toward the gap but do not, in the sources cited here, cross it — three from active inference and two from robust Bayes.

Five reviewed threads and... (part 2)

Thread 1 — Generative modeling and action (active inference). Discrete active inference is a mature,

Five reviewed threads and... (part 3)

synthesized formalism with standardized tensors and an expected-free-energy action rule (Friston 2010; Da Costa et al. 2020), and it is executable at scale through pymdp (Heins et al. 2022) and RxInfer (Bagaev et al. 2023).

Five reviewed threads and... (part 4)

Boundary: these sources do not by themselves equip the active-inference inference step with the contamination mechanism evaluated here.

Five reviewed threads and... (part 5)

Thread 2 — Collective belief coordination (active inference). Colonies coordinate by sharing beliefs and minimizing collective surprise, reproducing flocking (Heins et al. 2024), epistemic-community formation (Albarracin et al. 2022), and collective intelligence (Kaufmann et al. 2021).

Five reviewed threads and... (part 6)

Boundary: the cited collective-belief studies treat the broadcast posteriors as trusted and do not evaluate the robustness of fusion to misspecified or intentionally wrong members.

Five reviewed threads and... (part 7)

Thread 3 — Belief sharing and structure growth (active inference). Federated belief sharing fuses posteriors into a hive-mind consensus (Friston et al. 2024),

Five reviewed threads and... (part 8)

and post-hoc Bayesian model reduction (Friston and Penny 2011) together with optimal-design-flavored model selection (Smith et al. 2022) grow and prune the generative model. *Boundary:* in the cited belief-sharing line, fusion is exact-Bayes and trusting;

Five reviewed threads and... (part 9)

its connection to the robustness theory developed for federated learning is not evaluated.

Five reviewed threads and... (part 10)

Thread 4 — Federated and partitioned inference (robust Bayes). Federated learning aggregates decentralized models (McMahan et al. 2017), and partitioned variational inference gives the factor-algebra framework that unifies federated and continual learning (Bui et al. 2018; Ashman et al. 2022).

Five reviewed threads and... (part 11)

Boundary: the cited examples target parameter or predictive-model factors rather than the generative-model-bearing, action-selecting POMDP belief consensus used here.

Five reviewed threads and... (part 12)

Thread 5 — Generalized and robust Bayes (robust Bayes). Generalized Bayesian updating replaces the likelihood–KL pair with a loss–divergence pair (Bissiri et al. 2016; Knoblauch et al. 2022); bounded losses —

Five reviewed threads and... (part 13)

the density-power β-divergence (Basu et al. 1998) and generalized cross-entropy (Zhang and Sabuncu 2018) — deliver bounded influence; and FedGVI (Mildner et al. 2025) federates the robust objective with provable guarantees.

Five reviewed threads and... (part 14)

Boundary: the cited robust-Bayes apparatus does not evaluate active-inference POMDP belief consensus. Its behavior in the discrete categorical regime of the worked belief-sharing example (Friston et al. 2024) is the scoped setting evaluated here.

The belief-fusion bridge evaluated here

Across these reviewed threads, the missing intersection is specific. The cited active-inference sources provide belief fusion but not the contamination analysis used here. The cited robust-Bayes sources provide robust inference but not this acting-agent categorical consensus.

The belief-fusion bridge... (part 2)

We evaluate a bridge comprising three distinct robustness axes and one recovery anchor.

The belief-fusion bridge… (part 3)

The client axis is a FedGVI-faithful generalized-Bayes update using the β- and rcce-losses (eq. 7, eq. 8). The heuristic server axis reweights each agent by divergence from the current consensus (eq. 10).

The belief-fusion bridge... (part 4)

The objective-backed server axis is a conservative variational rule with a stated aggregation free energy and raw effective-weight bound (eq. 12).

The belief-fusion bridge… (part 5)

The shared anchor recovers the standard-Bayes client limit and project log-linear-pool server identity (sec. 9.1). Under the qualified bridge of sec. 8.8, the latter specializes Friston et al.'s Eq.

The belief-fusion bridge... (part 6)

7 message-combination term (Friston et al. 2024), not the complete source protocol. The algebra and executable residuals identify a recovery boundary, not an equivalence of literatures.

The belief-fusion bridge... (part 7)

Declared comparisons use paired Wilcoxon statistics (Wilcoxon 1945), Benjamini–Hochberg FDR (Benjamini and Hochberg 1995), percentile-bootstrap intervals (Efron and Tibshirani 1993), and observed-effect design-power planning. Source-bound reports make those analyses reproducible (Peng 2011);

The belief-fusion bridge... (part 8)

they do not promote a conditional simulation into a theorem.

Guarantee map: three robustness axes

A red-team review surfaced the paper's authoritative robustness taxonomy. Robustness enters in **three** places, with different claim owners, evidence classes, and permitted interpretations.

Client-side: source-theorem-backed

The per-agent generalized-Bayes update uses a bounded loss inside `generalized_posterior` (eq. 7, eq. 8). It is derived from eq. 2 and limits to NLL/Bayes as the loss parameter approaches zero (Corollary 6;

Client-side... (part 2)

Proposition 4). FedGVI's bounded-influence result transfers only under the source theorem's loss, model, and contamination assumptions (Mildner et al. 2025).

Server-side: heuristic

`robust_aggregate` discounts an agent by $\exp(-c\,\mathrm{KL}(q_n \,\|\, q))$. Its proved positive property is the recovery limit: at $c = 0$, it equals the standard log-linear pool (eq. 10, Theorem 5).

Server-side: heuristic (part 2)

It is not a closed-form FedGVI-objective minimizer and does not inherit the client-side bounded-influence result.

Server-side: objective-backed and conservative

`variational_aggregate` applies exact block updates that do not increase the stated free energy eq. 12.

Server-side... (part 2)

It recovers the log-linear pool in the trusting limit and bounds each raw effective weight by its base weight. Its declared tradeoff is a conservative consensus;

the objective-backed property does not imply peak-accuracy dominance.

Server-side... (part 3)

The robustness results (sec. 10.5) apply this map, and sec. 11.15 states the complete boundary. Effect sizes, intervals, and planning quantities characterize the server heuristic's conditional behavior;

Server-side... (part 4)

they do not transfer the client theorem or variational objective to that heuristic. Later sections refer back to this taxonomy rather than redefine it.